FAQ-23: How Do I Monitor Box With Dynatrace?

> **Series:** FAQ — Frequently Asked Questions | **Reference:** 23 — Monitoring Box With Dynatrace | **Created:** August 2026 | **Last Updated:** 10/29/2026

Overview

Box holds the documents. That is the whole reason this integration comes up: the audit trail of who opened, downloaded, shared, and deleted them is a compliance artifact, and it lives in a system nobody in the observability team administers. The request usually arrives phrased as a logging problem — "we need Box logs in Dynatrace" — and is answered as though the only open question were which ingestion endpoint to point at.

It is not. Two things about this integration are commonly assumed the wrong way round, and both change the design:

- **There is no turnkey Dynatrace integration for Box.** No Hub listing, no extension to activate. Verified three independent ways in August 2026 — Hub catalog search, web search, and the Dynatrace documentation assistant — all empty. You are building and owning this.
- **Box cannot push its audit feed to you.** Box webhooks exist, and they are the wrong tool: they attach to individual files and folders, cap at 1,000 resources per application, cannot be set on the root folder, and never carry logins, admin actions, or security alerts. The enterprise audit feed is available *only* by polling, and Box states plainly that it "does not support long polling." Every working design is a scheduled pull that owns a cursor.

That second point is the expensive one. A team that starts from "set up a webhook" will build a receiver, wire it correctly, and then discover that the events they actually needed were never eligible to arrive. This entry exists partly to prevent that specific week.

This entry is a **worked example of FAQ-19**. The generic decisions — signal classes, route selection, extract-then-discard, bucket isolation — are made there; here they are made concretely for Box, with a sizing example at **1,000,000 events per day** drawn from a real migration question.

> **A note on the DQL in this entry.** Box data will not exist in your tenant until the collector runs. The parsing, Shield and dedupe queries (§ 11, § 12.1–12.2) are therefore written to run standalone against a synthetic record using `data record(...)` — **zero bytes scanned** — so you can validate the shape before you have ingested anything. Swap `data record(...)`

> for `fetch logs | filter log.source == "box.events"` once data flows. The span query (§ 3.1) and the bucket, volume and staleness checks (§ 12.3, § 12.4, § 13.1) read real spans or logs and are billed as queries: each scanned more than a gigabyte on the validation tenant even with no Box data present (10/02/2026), so keep their timeframes short. All queries were executed against a live tenant on 08/25/2026; the changed ones again on 10/02/2026.

Table of Contents

1. Short Answer
2. What Box Is — and Why There Is No Turnkey Integration
3. The Four Signal Classes
4. The Events API — The Constraint That Shapes Everything
5. Sizing: A Worked Example at 1,000,000 Events/Day
6. Choosing the Collector — Workflow, Python or the Collector Receiver
7. Implementation A — Dynatrace Workflow
8. Implementation B — Python
9. Ingestion, Processing, and Storage
10. Field Mapping Starting Point
11. Box Shield Alerts
12. Querying Box Events in Grail
13. Operating the Collector
14. Availability Monitoring
15. Dashboards, Alerting, and SLOs
16. Implementation Sequence and Checklist
17. Common Gotchas
18. Recommended Approach

Prerequisites

Requirement	Details
Box licensing	Business plan or above for the Events API. **Box Shield is a separate SKU** — § 11 explains what its absence removes, and it is worth confirming before scoping

Requirement	Details
Box admin access	The service account must be an **enterprise admin or co-admin** holding *"Run new reports and access existing reports."* Without it the API returns nothing useful — which looks identical to "no events" (§ 17)
Box application	A Box app configured for **Client Credentials Grant** (server-side auth, no user interaction), with the application scope **Manage enterprise properties** checked, authorized in the Box Admin Console. Re-authorize the app in the Admin Console after changing its scopes
Dynatrace environment	SaaS with **Grail** and **OpenPipeline**
Outbound connectivity	If collecting via Workflow, `api.box.com` must be allowed under **Settings > General > External requests** (§ 7) — test one run after adding it
Security sign-off	This feed carries user identity, file names, and IP addresses. Decide retention and masking before building — see FAQ-19 § 4 and § 6
Related reading	**FAQ-19** (the generic pattern this instantiates), **FAQ-20** (the Zscaler worked example), **OPIPE** / **OPLOGS** (OpenPipeline depth), **ORGNZ** (buckets), **WFLOW** (workflows), **ALERT** / **SLO** / **DASH** (downstream)

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1. Short Answer

Build a scheduled collector that polls the Box Events API and posts to the Dynatrace Log Ingestion API. At typical enterprise volume — including **1,000,000 events per day** — a Dynatrace Workflow is sufficient and needs no external infrastructure.

The five decisions, made:

Decision	Answer	Why
Transport	Scheduled **pull** from `GET /events`	Box has no push for enterprise events. Webhooks cannot carry this feed (§ 2)
Stream	`admin_logs_streaming` for live; `admin_logs` for backfill	They are not interchangeable — different retention, ordering, and duplicate behaviour (§ 4)
Collector	**Workflow** up to ~35,000 events per run; Python beyond that or for deep backfill	The 120-second JS-runtime timeout is the only real boundary (§ 6)
Destination	Log Ingestion API v2, dedicated `log.source`	No events-per-minute cap; 10 MB / 50,000 records per request (§ 9)

Decision	Answer	Why
Retention	Extract metrics, then discard raw by default; dedicated bucket for what you keep	~1.5 GB/day at 1M events is a real budget line (§ 5)

Start with the signal that costs nothing. If your own applications call Box, OneAgent is already tracing those calls. Query them before building anything — see § 3.

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2. What Box Is — and Why There Is No Turnkey Integration

Box is a cloud content management platform — file storage, sharing, workflow, and governance — operated entirely by Box. In observability terms that single fact settles the architecture: there is no host to install OneAgent on, no process to inject, no infrastructure to discover. Everything you can learn about Box, you learn through its API.

This is the same shape as Zscaler (**FAQ-20**) and Adobe AEM as a Cloud Service (**FAQ-18**), and it is the pattern generalized in **FAQ-19**.

2.1 No Hub listing — verified, not assumed

As of August 2026 there is no Dynatrace Hub listing, extension, or first-party integration for Box. This was checked three independent ways — the Hub catalog, general web search, and the Dynatrace documentation assistant — and all three returned nothing.

Per the corpus rule that a negative claim needs a positive source: what is being reported here is the result of three catalog checks on 08/25/2026, not an inference from one empty search. Re-verify before quoting this in a design document; Hub listings appear over time.

The practical consequence is ownership. There is no vendor-maintained connector to inherit, and no upgrade path that will fix your integration for you. That is a normal position — it is simply better decided deliberately in week one than discovered in week three.

2.2 Why webhooks are the wrong answer — and why people reach for them

Box does publish webhooks, which is exactly why this goes wrong. They are a real feature that does a different job.

Box webhooks V2	Consequence for an audit feed
Attach to **specific files or folders**; **cannot be set on the root folder** (ID 0)	There is no account-wide subscription to create
Cap of **1,000 webhooks per application + user** combination	You could watch 1,000 folders — not an enterprise
Triggers are content actions (upload, copy, move, share, delete)	**No logins, no admin actions, no `SHIELD_ALERT`**
Delivery requires a 2xx within 30 s; Box retries up to 5× over 1 hour	A receiver outage drops events with no cursor to replay from
Token expiry degrades the payload to `NO_ACTIVE_SESSION`	Silent partial failure

Enterprise audit events are reachable only through `GET /events` with `stream_type` set to `admin_logs` or `admin_logs_streaming`, and Box's own documentation states that **"the enterprise event feed does not support long polling."** It is pull-only.

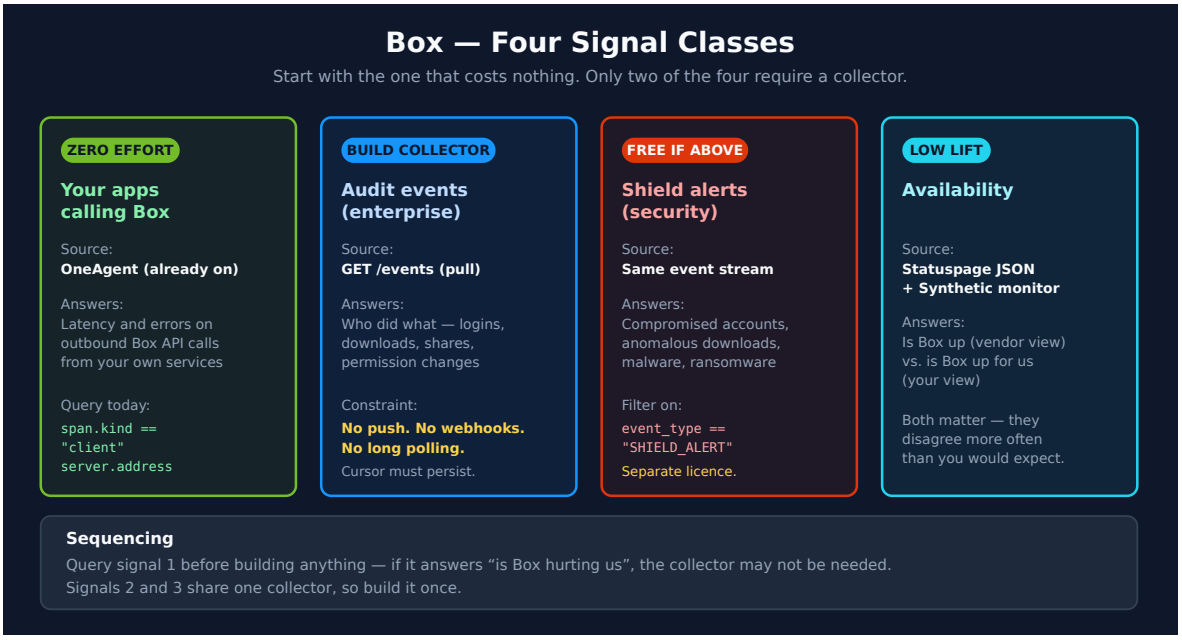
If someone has recommended a webhook for this, they have answered a different question. Webhooks are the right tool for reacting to changes in a specific high-value folder — a contract repository, a regulated dataset. They are structurally incapable of carrying the audit trail.

> **Sources:**
>
> - Webhooks (Box Dev Docs)
> - Webhooks V2 (Box Dev Docs) — *"Box retries webhook deliveries up to 5 times over a period of 1 hour. These numbers could be subject to change."*
> - Webhook limitations V2 (Box Dev Docs)
> - Get Enterprise Events (Box Dev Docs)
> - Box Hub catalog (Dynatrace)
> - **Catalog observation (08/25/2026):** § 2.1's no-listing claim is the result of three independent catalog checks on that date, not a single search

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3. The Four Signal Classes

FAQ-19 § 2 splits any third-party SaaS into signal classes and insists you choose which questions you are answering before choosing a transport. For Box there are four, and they differ enormously in cost.



3.1 Start with the signal that costs nothing

If any of your applications call the Box API, OneAgent is **already** capturing those as client spans. No Box configuration, no credentials, no collector. This is the fastest path to an answer and it is routinely skipped.

Run this before you build anything:

```
// Signal 1 — your own services calling Box. Requires no Box integration at all.
// Key on server.address: stable, and on every HTTP client span on the validation
tenant.
// Do not assume a URL field is populated. url.full is stable in the semantic
dictionary
// but was on 0 of 9.4M spans over 24 h on the validation tenant (10/15/2026);
http.url is
// not in the dictionary but was on 76,531 of them. Check countIf(isNotNull(...)) on
your
// own tenant before using either — see § 17. Same shape, different field:
span.status_code values are
// LOWERCASE, so `== "ERROR"` reports a 0% failure rate for every address.
// Divide a duration by 1ms to get a number — dividing by 1000000 still yields a
duration.
fetch spans, from:-24h
| filter span.kind == "client" and endsWith(server.address, "box.com")
| summarize {calls = count(),
             failures = countIf(span.status_code == "error"),
             p90_ms = percentile(duration, 90) / 1ms},
           by:{server.address}
| fieldsAdd failure_rate_pct = round(100.0 * failures / calls, decimals: 2)
| sort calls desc
```

If that returns rows, you already have Box performance telemetry and can build a dashboard today. If it returns nothing, confirm *why* before concluding your services do not call Box — an empty result here has at least three causes (no Box traffic, no OneAgent on the calling tier, or a mistyped field name), and only one of them is informative.

3.2 What each remaining signal is actually for

Signal	Answers	Does not answer
Audit events	Who accessed, shared, downloaded, or deleted what, when, from where	Whether Box was slow or erroring
Shield alerts	Which of those actions Box's own detection considers a threat	Anything Shield is not licensed to detect
Availability	Whether Box is reachable and healthy — vendor-declared and observed	Which user was affected, or what they were doing

The common mistake is expecting audit events to answer performance questions. They are an access record, not a latency record. Box does not publish per-request timing for its own service; if you need Box performance, it comes from signal 1 (your calls) or signal 4 (synthetics), never from the audit feed.

> **Sources:** Global field reference (DT docs), List user and enterprise events (Box Dev Docs). **Dictionary:** `server.address`, `url.full`, `url.path` (`stable`); no row for `http.url`, read 10/15/2026 (control: the same filter returned the other three). **Tenant observation (10/15/2026):** over 24 h, `url.full` was on 0 of 9.4M spans and `http.url` on 76,531; over 1 h, `server.address` and `url.path` were on all 57,594 HTTP client spans. An 08/25/2026 note here said `http.url` returned zero rows; that query was also filtered on Box addresses on a tenant with no Box traffic, so the filter produced the zero, not the field.

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4. The Events API — The Constraint That Shapes Everything

Everything enterprise-wide comes from one endpoint, `GET /events`, in one of two mutually exclusive modes. Choosing between them is the first real design decision, and most teams need both.

	`admin_logs_streaming`	`admin_logs`
Retention	**2 weeks**	**1 year**
Latency	Near real-time	Higher
Ordering	Out of chronological order	Chronological
Duplicates	**Yes** — dedupe on `event_id`	None
Designed for	Live monitoring and alerting	Historical query, backfill, investigation

Use streaming for the live pipeline and historical for backfill. They are not alternatives to choose between once; they are two jobs. A collector that only runs streaming has a two-week memory and no way to recover from a longer outage. A collector that only runs historical is too slow to alert on.

4.1 The constraints that shape the collector

Constraint	Value	Design consequence
No push	*"The enterprise event feed does not support long polling"*	Scheduled pull. There is no event-driven option
Page size	500 events maximum	Paging loop required; drives the sizing math in § 5
Cursor	`stream_position` in, `next_stream_position` out	**Must persist between runs** or you re-ingest or skip
Rate limit	1,000 requests/min per user; 429 returns `retry-after`	Exponential backoff. Rarely binding — see § 5
Permission	Enterprise admin/co-admin with *"Run new reports…"*, **and** the app's **Manage enterprise properties** scope	Missing the admin permission returns nothing, indistinguishable from no events. Missing the app scope is a separate cause — check both

4.2 The cursor is the whole design

Everything difficult about this collector reduces to one question: *where did I get to last time?* Box hands you `next_stream_position` with every page; you hand it back on the next call. Lose it and you either replay events you already have (paying twice) or skip events you never got (a compliance gap you cannot detect after the fact).

Because `admin_logs_streaming` retains only two weeks, a lost cursor is not always recoverable. This is the single highest-consequence property of the integration and it drives § 13.

> **Sources:**
>
> - Get Enterprise Events (Box Dev Docs)
> - List user and enterprise events (Box Dev Docs) — *"The user making the API call will need to have admin privileges, and the application will need to have the scope manage enterprise properties checked."*
> - Platform app approval (Box Dev Docs) — *"When an application's scopes or access level change, the application must be re-authorized for the changes to take effect."*
> - Rate Limits (Box Dev Docs)
> - New Enterprise Event Stream API (Box Support)

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5. Sizing: A Worked Example at 1,000,000 Events/Day

This section works a real number end to end, because the volume question is usually asked about the wrong endpoint. A migrating customer asked whether a webhook could handle 1M Box events per day; the answer they received addressed only the Dynatrace ingest limits. Both halves matter, and — as it turns out — **neither is the constraint.**

5.1 Both endpoints have ample headroom

Side	Documented limit	At 1M events/day	Verdict
Dynatrace ingest	**No events-per-minute cap**; 10 MB and 50,000 records per request	~11.6 events/sec	Not a constraint
Box Events API	500 events/page; 1,000 requests/min per user	1M ÷ 500 = 2,000 pages/day ≈ **1.4 req/min**	~0.1% of budget

1,000,000 events per day is a modest load for both systems. **The bottleneck is neither endpoint — it is that no collector exists between them.** That is the part a question about "which ingestion option handles the volume" does not surface, because the ingest API is the destination, not the connection.

5.2 What that means for a Workflow

Polling every 5 minutes gives 288 runs/day and ~3,472 events per run — about **7 pages of 500**. Against a 120-second execution ceiling that is comfortable:

Quantity	Value	Basis
Runs per day (5-min schedule)	288	Schedule
Events per run	~3,472	1M ÷ 288
Pages per run	~7	500 events/page
Estimated time per run	~15 s	~1–2 s per page — **an estimate, not a documented figure**
Execution ceiling	**120 s**	JS-runtime hard limit
Approximate headroom used	**~10%**	Derived

Mark the page-latency assumption clearly and measure it in your own tenant. The documented numbers here — 120 s, 500 events/page, 1,000 req/min, no ingest rate cap — are verified. The per-page round-trip time is not, and it is the one input that could move the conclusion.

5.3 The cost line nobody quotes

At roughly 1.5 KB per event:

	Volume
Per day	~1.5 GB
Per month	**~45 GB**

That is a real DPS line item, and it arrives whether or not anyone budgeted for it. It is also the strongest argument for the extract-then-discard posture in § 9: most of those 45 GB are `PREVIEW` and `DOWNLOAD` events whose analytical value is fully captured by a counter.

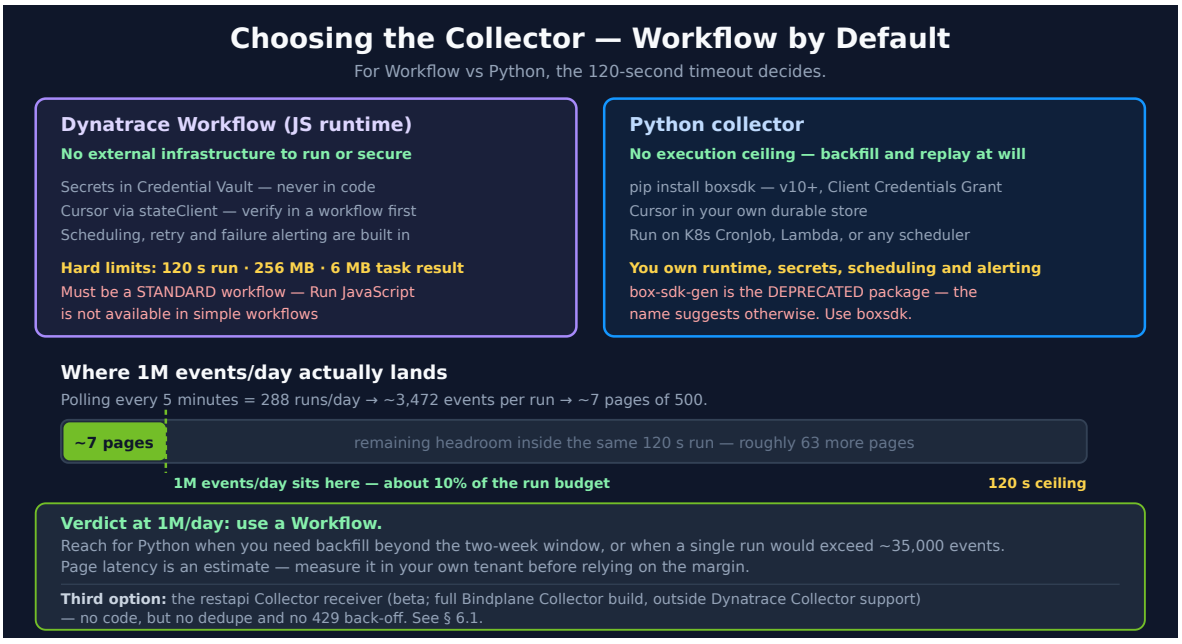
Decide retention *before* you turn the collector on. Reducing retention later does not refund what you already ingested.

> **Sources:** Log ingestion limits (DT docs), Rate Limits (Box Dev Docs), Get Enterprise Events (Box Dev Docs).
>
> **Derived:** § 5.1–5.3 combine the documented per-side limits with a 1M/day rate and a ~1.5 KB average event size; the per-page latency and event size are estimates to verify locally.

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6. Choosing the Collector — Workflow, Python or the

Collector Receiver



	Workflow (JS runtime)	Python
Infrastructure	None — runs inside Dynatrace	You provision, run, patch, secure
Execution ceiling	**120 s** per run (hard)	None
Memory / task result	256 MB / 6 MB task result	Yours
Secrets	Credential Vault	Your own secret store
Cursor	`stateClient` app state — verify it works from a workflow first (§ 7.1)	Your own durable store
Scheduling, retry, failure alerting	Built in	You build all three
Backfill over months	Impractical	Natural fit

At 1M events/day, use a Workflow. It clears the ceiling with roughly 90% headroom (§ 5.2) and removes an entire class of operational work — there is no container to keep patched, no secret to rotate outside Dynatrace, and failure alerting is already there.

Reach for Python when:

- You need to backfill more than the two-week streaming window, or replay a year of `admin_logs`.
- Sustained volume pushes a single run past roughly 35,000 events (~70 pages).
- Policy requires the collector to live in your own estate.

- You want to enrich against systems the JS runtime cannot reach.

A common and sensible split: **Python for the one-time historical backfill, Workflow for steady state.** They write to the same `log.source` and the same bucket, and the backfill runs once.

> **Sources:** JavaScript runtime limits (Dynatrace Developer), Run JavaScript action (DT docs) — *"The task result size is limited to 6 MB."*, Get Enterprise Events (Box Dev Docs). **Derived:** the ~35,000-events-per-run boundary combines the documented 120 s ceiling with the 500-event page size and an estimated per-page latency — verify locally per § 5.2.

6.1 The REST API Collector receiver

A third option needs no code of your own. The `restapi` receiver for the OpenTelemetry Collector is a generic REST poller: it calls a URL on a schedule, follows the API's pagination, and emits each item of the response array as a log record. It supports OAuth 2.0 client credentials with extra token parameters, which is what Box's enterprise service account needs, and an opaque cursor such as `stream_position`. It is maintained in the `dynatrace-bindplane-otel-contrib` repository and built into the full Dynatrace Bindplane Distribution of the Collector (DBDOT). It is not in the slim DBDOT build or in the Dynatrace OTel Collector distribution, and Dynatrace's Collector support covers only the components that distribution includes. Its README configures it in plain Collector YAML, so Bindplane is optional. Its stability is **beta**.

Measured against the rules this entry depends on, v1.2.0 (released 10/28/2026) gets the cursor rule right and falls short in several other places:

Rule from this entry	**`restapi` receiver v1.2.0**	**Effect**
Write the cursor only after ingest succeeds (§ 7.3, gotcha 5)	Each page goes to the pipeline before the cursor advances. A pipeline error ends the poll without saving	Meets the rule as far as the exporter — see the queue row
Persist the cursor between runs (§ 4.2)	Saved after each page and at the end of each poll, but only with a `storage` extension. Changing `url` invalidates the saved cursor	Without `storage`, a restart is a first run (next row)

Rule from this entry	`restapi` receiver v1.2.0	Effect
Start a new stream at `now` (§ 8.2)	An opaque cursor cannot be seeded: `starting_offset` is rejected with `offset_type: opaque`, and the first request omits `stream_position`. Adding `stream_position=now` to the URL does not help — the receiver adds its own parameter beside it on later requests	Box documents `now` and *"`0` or `null` will return all events"*, not an omitted parameter. If the first run starts at the beginning, it replays up to two weeks, and records older than the Log Age limit are dropped at ingest
Back off on `429` using `retry-after` (§ 4.1)	No `429` or `Retry-After` handling. Any non-2xx response is an error, and the next attempt waits for the next poll, whose interval grows toward `max_poll_interval`	The cursor does not move, so nothing is lost. Rarely binding at Box's rate limit (§ 5)
No `timestamp` on events older than ~23 h (§ 9.1, gotcha 16)	The record time comes from the first top-level `timestamp`, `time`, `created_at` … field — for Box, `created_at`, always	A backlog older than the Log Age limit is discarded at OTLP ingest. Applying the 23 h rule needs a `transform` processor rule (not written or tested here)
Dedupe on `event_id` (§ 4, § 12.2)	No deduplication	`admin_logs_streaming` duplicates reach Dynatrace. Count them with § 12.2 and dedupe at query time
A crash must not lose collected events (§ 13)	The exporter's sending queue is in memory unless you give it `storage`, so a page counts as forwarded once it is queued	A crash can lose queued records the cursor has already passed. Set `sending_queue.storage` too
Poll every 5 minutes (§ 5)	Adaptive interval: it drops to `min_poll_interval` only when `page_limit` is set and the last page was full — not whenever data arrives, as the README says. Without `page_limit` each poll drains every page and the interval settles at `max_poll_interval`	Set `max_poll_interval: 5m` (the default) for the § 5 cadence

If you test it, start from this configuration — it has not been run against a Box enterprise:

```
extensions:
  file_storage:
    directory: /var/lib/otelcol/box   # must survive restarts
receivers:
  restapi/box:
    url: https://api.box.com/2.0/events?stream_type=admin_logs_streaming
    auth_mode: oauth2
    oauth2:
      client_id: ${env:BOX_CLIENT_ID}
      client_secret: ${env:BOX_CLIENT_SECRET}
      token_url: https://api.box.com/oauth2/token
      endpoint_params:
        box_subject_type: enterprise
        box_subject_id: ${env:BOX_ENTERPRISE_ID}
    response_field: entries
    pagination:
      mode: offset_limit
      offset_limit:
        offset_field_name: stream_position
        next_offset_field_name: next_stream_position
        offset_type: opaque
        limit_field_name: limit
        limit: 500
    max_poll_interval: 5m
    storage: file_storage
exporters:
  otlphttp:
    # your environment's OTLP endpoint and token
    sending_queue:
      storage: file_storage   # persistent queue
```

Add `file_storage` to `service.extensions` and the receiver to a logs pipeline that uses the exporter. The receiver puts the whole Box event in the log body. The § 10 field mapping and the § 12 and § 13 queries read the `box.*` attributes and the `log.source` the Workflow and Python collectors write, so map them in OpenPipeline or adapt the queries before you rely on them.

When it fits: steady-state streaming, once you have run its first poll against a Box sandbox and seen where the stream starts. The Workflow (§ 7) stays the default, and the `admin_logs` backfill (§ 8.3) is unchanged: the receiver stamps every record with its event time, so without a `transform` rule it cannot replay events older than the Log Age limit. The receiver is new and actively developed, so re-check this table against each release.

> **Sources:**
>
> - restapireceiver README (Dynatrace GitHub) — *"When a storage extension is configured, the receiver saves its pagination state to storage after each page fetch."*
> - restapireceiver metadata (Dynatrace GitHub) — *"beta: [logs, metrics]"*
> - OpenTelemetry Collector (DT docs) — *"This support covers only the components and their versions that are*

> *included in the Dynatrace OTel Collector."*
>
> - List user and enterprise events (Box Dev Docs) — *" `now` will return an empty list events and the latest stream position for initialization."*
> - Client credentials grant (Box Dev Docs) — *"Every token request names one of three subjects."*
> - Exporter helper (OpenTelemetry GitHub) — *"There is no in-memory queue when set."*
> - Ingest OTLP logs (DT docs) — *"Log events older than the Log Age limit are discarded."*
> - **Code read (10/29/2026):** at tag `v1.2.0`, `receiver.go` sends each page to the pipeline before reading the next cursor and returns on a pipeline error before saving; `client.go` treats every non-2xx status as an error and contains no `Retry-After` handling (control: its OAuth2 `token_url`, `scopes` and `endpoint_params` handling is found in the same file); `client.go` adds pagination parameters to any already in the URL rather than replacing them; `config.go` rejects `starting_offset` with `offset_type: opaque`; `pagination.go` omits the cursor parameter until a token arrives; `logs.go` reads the record time from the first of `timestamp`, `time`, `created_at`, `createdAt`, `date`, `datetime`, `@timestamp`. Component lists: `restapireceiver v1.2.0` is in the full DBDOT manifest and in neither the slim DBDOT manifest nor the Dynatrace OTel Collector manifest (0.59.0). Not run against Box
> - **Derived:** the queue row combines the exporter-helper sentence with the receiver forwarding a page before it saves the cursor

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7. Implementation A — Dynatrace Workflow

The recommended path at typical volume. No infrastructure, secrets stay in the Credential Vault, and scheduling plus failure alerting come free.

7.1 Three things to set up first

Step	What	Why it matters
1. Credential Vault entry	Store the Box `client_id` / `client_secret` as a username/password credential with the **AppEngine** scope selected, **Allow access without app context** turned on, and access granted to the **workflow actor**	Never put Box secrets in workflow code — they are readable by anyone who can view the workflow. Workflow JavaScript tasks run without an app context, so a credential missing any of the three settings cannot be read by the task

Step	What	Why it matters
2. Outbound allowlist	Add `api.box.com` — and `status.box.com` if you poll it (§ 14.1) — under **Settings > General > External requests** (*New host pattern*)	Every HTTP call from the task is checked against this list. No page documents what a blocked call returns: in the collector below an uncaught error fails the task, while code that catches errors turns it into an empty run. Run the workflow once after adding the host and check that events arrive
3. Standard workflow	Create a **standard** workflow, not a simple one	A simple workflow cannot contain a **Run JavaScript** action at all. Standard workflows also bill differently — see ALERT-03 § 1

> **Verify the cursor store before you schedule the collector.** The collector keeps its cursor in app state through `stateClient`. Two Dynatrace pages leave open whether app state is reachable from a Run JavaScript task: the state service says *"These states are only accessible from their respective apps and can't be accessed using a different app or ad-hoc functions."*, and the Run JavaScript page says *"JavaScript tasks run without an app context."* The code below matches the `@dynatrace-sdk/client-state` typings, but it has not been run in a workflow. Before relying on it, build a two-task test workflow: the first task calls `setAppState`, the second calls `getAppState` and returns the value. The workflow actor needs `state:app-states:read` and `state:app-states:write` (the scopes the SDK documents for these calls). If the value does not come back, keep the cursor in a store you have confirmed the workflow can read and write — the rest of the design does not change. Without a working cursor the run either fails or starts at `"now"` every time and never collects the events between runs.

7.2 The collector

A single **Run JavaScript** action on a 5-minute schedule. It authenticates, reads the cursor, pages until drained, ingests, and writes the cursor back.

```
import { credentialVaultClient, logsClient } from "@dynatrace-sdk/client-classic-
environment-v2";
import { stateClient, isNotFound } from "@dynatrace-sdk/client-state";

const BOX_CREDENTIAL_ID = "CREDENTIALS_VAULT-XXXXXXXXXXXXXXXX";
const BOX_ENTERPRISE_ID = "1234567";
const CURSOR_KEY        = "box-events-cursor";
const LOG_SOURCE        = "box.events";
const PAGE_LIMIT        = 500;
const MAX_PAGES         = 60;      // stay clear of the 120 s ceiling
const INGEST_BATCH      = 1000;    // well under 50,000 records / 10 MB
const MAX_EVENT_AGE_MS  = 23 * 60 * 60 * 1000; // log ingest drops timestamps &gt; 24
h old (§ 9.1)

export default async function () {
  // --- 1. Authenticate (Client Credentials Grant) -------------------------
  const creds = await credentialVaultClient.getCredentialsDetails({
    id: BOX_CREDENTIAL_ID,
  });

  const tokenRes = await fetch("https://api.box.com/oauth2/token", {
    method: "POST",
    headers: { "Content-Type": "application/x-www-form-urlencoded" },
    body: new URLSearchParams({
      grant_type: "client_credentials",
      client_id: creds.username,
      client_secret: creds.password,
      box_subject_type: "enterprise",
      box_subject_id: BOX_ENTERPRISE_ID,
    }).toString(),
  });

  if (!tokenRes.ok) {
    // Fail loudly. A silent auth failure looks exactly like "no events".
    throw new Error(`Box auth failed: ${tokenRes.status} ${await tokenRes.text()}`);
  }
  const { access_token } = await tokenRes.json();

  // --- 2. Read the cursor -------------------------------------------------
  // "now" on first run starts at the live edge. Use 0 to replay everything
  // the streaming window still holds (up to two weeks).
  let streamPosition = "now";
  try {
    const saved = await stateClient.getAppState({ key: CURSOR_KEY });
    if (saved?.value) streamPosition = saved.value;
  } catch (e) {
    // No cursor yet — first run. Any other error should surface.
    // The SDK error has no top-level `status`; use the exported type guard.
    if (!isNotFound(e)) throw e;
  }

  // --- 3. Page until drained ----------------------------------------------
```

```
  const seen = new Set();
  const records = [];
  let pages = 0;

  while (pages &lt; MAX_PAGES) {
    const url =
      `https://api.box.com/2.0/events?stream_type=admin_logs_streaming` +

`&amp;limit=${PAGE_LIMIT}&amp;stream_position=${encodeURIComponent(streamPosition)}`;

    const res = await fetch(url, {
      headers: { Authorization: `Bearer ${access_token}` },
    });

    if (res.status === 429) {
      // Respect Box's backoff and finish on the next scheduled run.
      break;
    }
    if (!res.ok) {
      throw new Error(`Box events failed: ${res.status} ${await res.text()}`);
    }

    const page = await res.json();
    const entries = page.entries ?? [];

    for (const evt of entries) {
      // admin_logs_streaming can repeat events — dedupe within the run.
      if (seen.has(evt.event_id)) continue;
      seen.add(evt.event_id);

      const record = {
        content: JSON.stringify(evt),
        "log.source": LOG_SOURCE,
        "box.created_at": evt.created_at,
        "box.event_id": evt.event_id,
        "box.event_type": evt.event_type,
        // SHIELD_ALERT events carry an anonymous created_by (empty login);
        // the affected account is in additional_details.shield_alert.user (§ 11.2).
        "box.user.login": evt.created_by?.login ||
evt.additional_details?.shield_alert?.user?.email,
        "box.item.name": evt.source?.item_name,
        "box.ip_address": evt.ip_address,
      };
      // Log ingest DROPS a record whose timestamp is older than 24 h. For a
      // backlog (first run from 0, a long outage), leave timestamp unset so the
      // record takes its ingest time, and query event time on box.created_at.
      if (Date.now() - Date.parse(evt.created_at) &lt; MAX_EVENT_AGE_MS) {
        record.timestamp = evt.created_at;
      }
      records.push(record);
    }
```

```
    streamPosition = page.next_stream_position;
    pages++;

    // A short page usually means the live edge — but Box can return fewer than
    // `limit` while events remain. Stopping here only defers them to the next
    // run; the cursor still points at them, so nothing is lost.
    if (entries.length &lt; PAGE_LIMIT) break;
  }

  // --- 4. Ingest in batches ---------------------------------------------
  for (let i = 0; i &lt; records.length; i += INGEST_BATCH) {
    await logsClient.storeLog({
      type: "application/json; charset=utf-8",
      body: records.slice(i, i + INGEST_BATCH),
    });
  }

  // --- 5. Persist the cursor LAST ---------------------------------------
  // Only after a successful ingest. Writing it earlier means a failed ingest
  // silently loses every event in this batch.
  await stateClient.setAppState({
    key: CURSOR_KEY,
    body: { value: String(streamPosition) },
  });

  return { pages, ingested: records.length, cursor: streamPosition };
}
```

7.3 Six details in that code that are load-bearing

Detail	Why
Cursor written last	If ingest fails after the cursor advances, those events are skipped by the live collector — and once they fall out of the two-week streaming window, only an `admin_logs` backfill (§ 13.3) can recover them. Write-after-success makes a failed run replay instead of lose
No TTL on the cursor	`stateClient` TTL (`validUntilTime`) accepts now+1m … now+90d. Set one and your cursor silently expires, resetting the collector. **Omit it**
`MAX_PAGES` guard	Bounds the run below the 120 s ceiling. A backlog drains across several runs rather than timing out forever on the first
Throwing on auth failure	An unhandled non-OK response would leave you with an empty ingest that looks identical to a quiet Box tenant
`isNotFound(e)` on the first cursor read	The SDK's error object carries the HTTP response under `response`, not a top-level `status` — a `e.status !== 404` test is always true and makes every first run throw

Detail	Why
No `timestamp` on events older than ~23 h	Log ingest drops any record timestamped more than 24 h in the past (§ 9.1). Replaying a backlog with `timestamp: created_at` loses it silently; ingest-time stamping plus `box.created_at` keeps it

7.4 Alert on the collector, not just on Box

A Workflow that stops running produces no logs — and no logs looks exactly like no activity. Add a second workflow that runs the § 13 staleness query and pages someone when it goes stale. **This is not optional at a two-week retention window.**

> **Sources:**
>
> - JavaScript runtime limits (Dynatrace Developer)
> - client-state SDK (Dynatrace Developer) — exports the `isNotFound` type guard (checked in `@dynatrace-sdk/client-state` 1.11.0 typings, 09/28/2026)
> - Manage secrets (Dynatrace Developer) — *"Ensure to use the AppEngine scope for your credentials."*
> - Run JavaScript action (DT docs) — *"The following credential settings are required. The AppEngine scope is selected. Allow access without app context is turned on. The workflow actor has access to the credential."*; *"All HTTP calls are validated against the global allowlist."*
> - State service (Dynatrace Developer) — *"These states are only accessible from their respective apps and can't be accessed using a different app or ad-hoc functions."*
> - Create a simple workflow (DT docs) — *"You can use all available actions to create a simple workflow except Run JavaScript"*
> - Credential vault client (Dynatrace Developer)
> - Client Credentials Grant (Box Dev Docs)
> - List user and enterprise events (Box Dev Docs)
> - **Derived:** § 7.3's write-cursor-last rule combines the documented two-week streaming retention with the collector resuming from its stored cursor — events behind an advanced cursor are reachable again only through an `admin_logs` backfill (§ 13.3)

---

8. Implementation B — Python

Use Python for historical backfill, for replay, or when sustained volume exceeds what a single 120-second run can drain.

8.1 Install the right SDK — this one is easy to get backwards

```
pip install boxsdk requests
```

`boxsdk` (v10+) is the current Box Python SDK. `box-sdk-gen` is the deprecated one. The naming is actively misleading — "gen" sounds like the newer generation, and it was, briefly: it shipped as a standalone package and was deprecated on **17 September 2025** when its functionality was absorbed into the core SDK at v10.

The PyPI release history settles it: `boxsdk` 10.17.0 shipped 1 October 2026, while `box-sdk-gen` has not released since 05 September 2025 (both re-checked 10/02/2026).

The second half of the trap: you install `boxsdk`, but you import `box_sdk_gen`. From v10 the `boxsdk` distribution contains only the generated package — its PyPI page says v10 *"fully and exclusively replaces the old `boxsdk` package"* — so `from boxsdk import Client, CCGAuth`, which every pre-v10 example uses, fails with `ModuleNotFoundError`. The collector below imports `BoxClient`, `BoxCCGAuth` and `CCGConfig` from `box_sdk_gen`.

8.2 The collector

Syntax- and import-checked against `boxsdk` 10.17.0 (10/02/2026), including the `client.events.get_events` signature; it has not been run end-to-end against a Box enterprise. It authenticates to Dynatrace with a **platform token** (scope `openpipeline:logs:ingest`), because Latest Dynatrace environments have no classic access tokens. On a classic or hybrid environment a classic access token with `logs.ingest` also works — send `Api-Token <token>` instead of `Bearer <token>`.

```
"""Poll the Box enterprise event stream and ship to Dynatrace.

Steady state:  --stream-type admin_logs_streaming
Backfill:      --stream-type admin_logs --created-after 2026-01-01T00:00:00Z \
               --cursor-file /var/lib/box/cursor-backfill

Requires Python 3.11+ (datetime.fromisoformat accepts Box's offsets and "Z").
"""
import argparse
import json
import os
import time
from datetime import datetime, timedelta, timezone
from pathlib import Path

import requests
from box_sdk_gen import BoxCCGAuth, BoxClient, CCGConfig
from box_sdk_gen.managers.events import GetEventsStreamType

BOX_CLIENT_ID     = os.environ["BOX_CLIENT_ID"]
BOX_CLIENT_SECRET = os.environ["BOX_CLIENT_SECRET"]
BOX_ENTERPRISE_ID = os.environ["BOX_ENTERPRISE_ID"]

DT_ENV_URL = os.environ["DT_ENV_URL"]        # https://abc12345.live.dynatrace.com
DT_PLATFORM_TOKEN = os.environ["DT_PLATFORM_TOKEN"]  # scope:
openpipeline:logs:ingest

LOG_SOURCE    = "box.events"
PAGE_LIMIT    = 500
INGEST_BATCH  = 1000     # well under the 50,000-record / 10 MB request limits
MAX_EVENT_AGE = timedelta(hours=23)  # ingest drops timestamps &gt; 24 h old (§ 9.1)


def box_client() -&gt; BoxClient:
    """Server-side auth. No user interaction, no refresh-token juggling."""
    config = CCGConfig(
        client_id=BOX_CLIENT_ID,
        client_secret=BOX_CLIENT_SECRET,
        enterprise_id=BOX_ENTERPRISE_ID,
    )
    return BoxClient(auth=BoxCCGAuth(config=config))


def read_cursor(cursor_file: Path, default: str) -&gt; str:
    return cursor_file.read_text().strip() if cursor_file.exists() else default


def write_cursor(cursor_file: Path, position: str) -&gt; None:
    """Write atomically — a torn cursor file is worse than no cursor file."""
    cursor_file.parent.mkdir(parents=True, exist_ok=True)
    tmp = cursor_file.with_suffix(".tmp")
    tmp.write_text(str(position))
```

```
    tmp.replace(cursor_file)


def to_dt_record(evt: dict) -&gt; dict:
    """Flatten a Box event into a Dynatrace log record."""
    created_by = evt.get("created_by") or {}
    source     = evt.get("source") or {}
    created_at = evt.get("created_at")
    details    = evt.get("additional_details")
    shield     = (details.get("shield_alert") if isinstance(details, dict) else None)
or {}
    record = {
        "content":         json.dumps(evt, separators=(",", ":")),
        "log.source":      LOG_SOURCE,
        "box.created_at":  created_at,
        "box.event_id":    evt.get("event_id"),
        "box.event_type":  evt.get("event_type"),
        # SHIELD_ALERT: created_by is anonymous; the account is shield_alert.user.
        "box.user.login":  created_by.get("login") or (shield.get("user") or
{}).get("email"),
        "box.item.name":   source.get("item_name"),
        # File/folder sources use item_type/item_id; user sources use type/id.
        "box.item.type":   source.get("item_type") or source.get("type"),
        "box.ip_address":  evt.get("ip_address"),
    }
    # Only stamp event time when ingest will accept it. An older event keeps
    # its ingest time as timestamp; query its real time on box.created_at.
    if created_at:
        age = datetime.now(timezone.utc) - datetime.fromisoformat(created_at)
        if age &lt; MAX_EVENT_AGE:
            record["timestamp"] = created_at
    return record


def ingest(records: list[dict]) -&gt; None:
    """POST to the Dynatrace Log Ingestion API, batched."""
    url = f"{DT_ENV_URL}/api/v2/logs/ingest"
    headers = {
        "Authorization": f"Bearer {DT_PLATFORM_TOKEN}",
        "Content-Type": "application/json; charset=utf-8",
    }
    for i in range(0, len(records), INGEST_BATCH):
        batch = records[i:i + INGEST_BATCH]
        for attempt in range(5):
            resp = requests.post(url, headers=headers,
                                 data=json.dumps(batch), timeout=30)
            if resp.status_code == 204:
                break
            # 200 is a PARTIAL success: some records were dropped (for example
            # "... events were not ingested because of timestamp out of correct
            # time range"). Treat it as a failure so the cursor does not advance.
            if resp.status_code == 200:
```

```
                raise RuntimeError(f"Dynatrace partial ingest: {resp.text}")
            # 413 means the batch expanded past 10 MB (or 16 MB after
            # preprocessing) — halving is the fix, not retrying.
            if resp.status_code == 413 and len(batch) &gt; 1:
                mid = len(batch) // 2
                ingest(batch[:mid])
                ingest(batch[mid:])
                break
            if resp.status_code in (429, 500, 502, 503, 504):
                time.sleep(2 ** attempt)
                continue
            resp.raise_for_status()
            raise RuntimeError(f"Unexpected ingest response {resp.status_code}")
        else:
            raise RuntimeError("Dynatrace ingest failed after 5 attempts")


def collect(stream_type: str, cursor_file: Path,
            created_after: datetime | None = None) -&gt; int:
    client = box_client()
    # "now" returns no events, only the live-edge position; "0" replays
    # everything the stream still holds. Backfill wants the latter.
    default = "0" if stream_type == "admin_logs" else "now"
    position = read_cursor(cursor_file, default)
    seen: set[str] = set()
    total = 0

    while True:
        # The SDK handles auth, retry and token refresh for us.
        page = client.events.get_events(
            stream_type=GetEventsStreamType(stream_type),
            stream_position=position,
            limit=PAGE_LIMIT,
            created_after=created_after,
        )
        entries = page.entries or []

        records = []
        for evt in entries:
            data = evt.to_dict()
            eid = data.get("event_id")
            # admin_logs_streaming repeats events; admin_logs does not.
            if eid in seen:
                continue
            seen.add(eid)
            records.append(to_dt_record(data))

        if records:
            ingest(records)
            total += len(records)

        # Persist only after a successful ingest — and on an EMPTY page too,
```

```
        # or a cursor that started at "now" is never initialized.
        position = str(page.next_stream_position)
        write_cursor(cursor_file, position)

        # Empty means the live edge. A short page may not be — Box can return
        # fewer than `limit` while events remain — but stopping only defers
        # them to the next run; the saved cursor still points at them.
        if len(entries) &lt; PAGE_LIMIT:
            break

    return total


if __name__ == "__main__":
    ap = argparse.ArgumentParser()
    ap.add_argument("--stream-type", default="admin_logs_streaming",
                    choices=["admin_logs_streaming", "admin_logs"])
    ap.add_argument("--created-after", default=None,
                    help="RFC3339 timestamp; admin_logs backfill only")
    ap.add_argument("--cursor-file", type=Path,
                    default=Path(os.environ.get("BOX_CURSOR_FILE",
                                                "/var/lib/box/cursor")),
                    help="use a separate file per stream type")
    args = ap.parse_args()

    after = (datetime.fromisoformat(args.created_after)
             if args.created_after else None)
    count = collect(args.stream_type, args.cursor_file, after)
    print(f"ingested {count} events")
```

8.3 Running it

Mode	Command	Notes
Steady state	`python box_collector.py`	Every 5 min via cron, K8s `CronJob`, or a systemd timer
Backfill	`python box_collector.py --stream-type admin_logs --created-after 2026-01-01T00:00:00Z --cursor-file /var/lib/box/cursor-backfill`	Run once. The **separate cursor file** keeps it from clobbering the live cursor. Events older than 24 h land with their ingest time as `timestamp` — query them on `box.created_at` (§ 9.1)

Keep the two cursors apart. A backfill sharing the live cursor will drag the streaming collector backwards or forwards unpredictably — the most common way this design corrupts itself.

8.4 Where to run it

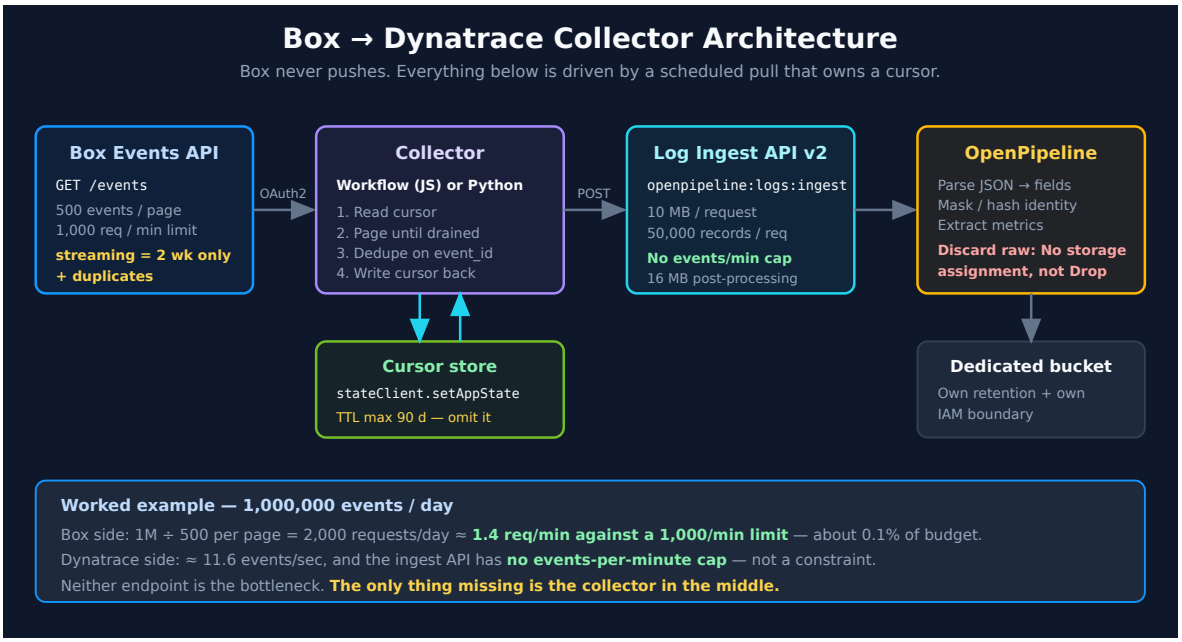
Host	Fit
Kubernetes `CronJob`	Best fit if you already run K8s — secrets, scheduling, and restarts are solved
AWS Lambda / Azure Functions	Fine; keep the cursor in DynamoDB, Table Storage, or a parameter store rather than a file
A VM with cron	Works. Ensure the cursor file is on persistent storage, not ephemeral disk

Whatever runs it needs monitoring of its own — see § 13.

> **Sources:**
>
> - boxsdk on PyPI — *"Starting with v10, the SDK is built entirely on the generated `box_sdk_gen` package, which fully and exclusively replaces the old `boxsdk` package."*
> - Deprecated Box Next Gen Python SDK (Box Dev Docs)
> - Client Credentials Grant (Box Dev Docs)
> - POST ingest logs (DT docs) — *"Platform Token / OAuth: One of the following scopes is required: openpipeline:logs:ingest storage:logs:write (legacy)"*
> - Upgrade from classic access tokens (DT docs) — *"Classic access tokens don't exist in latest environments, and v2/apiTokens isn't available."*
> - Log ingestion limits (DT docs) — *"200 - in case some of the events in the payload have timestamps earlier than the current time minus 24 hours."*
> - List user and enterprise events (Box Dev Docs) — *" `now` will return an empty list events and the latest stream position for initialization"*; *"Sometimes, the events less than the limit requested can be returned even when there may be more events remaining."*
> - **Derived:** § 8.1's current-vs-deprecated verdict comes from PyPI release dates (re-checked 10/02/2026) against Box's deprecation notice

---

9. Ingestion, Processing, and Storage



9.1 Ingest limits that actually bind

Limit	Value	Consequence
Events per minute	**No limit**	Volume is not the problem (§ 5)
Payload per request	**10 MB**	Batch by size, not only by count
Records per request	**50,000**	Rarely the binding limit — size hits first
Post-processing expansion	Rejected above **16 MB**	A batch inside 10 MB can still fail after processing. Halve on 413
Attributes per record	500	Ample for Box events
Attribute value length	32 kB	A large `additional_details` can approach this
Log age	**24 h** — 72 h from SaaS 1.348 (staged tenant rollout)	A record timestamped earlier is **dropped**; a mixed batch returns **200** with a "not ingested" message, not an error. Backfill must not stamp old events with their event time (§ 7.3, § 8.2)

9.2 Extract, then discard — in that order

The mechanics and the trap are in **FAQ-19 § 4**; do not configure this without reading it. In short:

Extract	Example	Retention posture
Metric	Event counts by `event_type` and user; download volume by user; failed-login counts	**Keep the metric.** Discard the source record unless separately justified
Business event	A file in a regulated folder was shared externally	**Keep**, minimal fields
Alert	`SHIELD_ALERT` (§ 11)	**Keep** with full context
Raw record	Every `PREVIEW` and `DOWNLOAD` event	**Discard by default.** This is the bulk of your 45 GB/month

> **The trap, stated once more because it silently defeats the whole design:** `Drop record` in the **Processing** stage pre-empts every extractor downstream — your metrics never get built. Use `No storage assignment` in the **Bucket assignment** stage instead: the record still runs through metric extraction, which comes after it, and is simply not stored. Extract first, discard last.

Field-level controls in the Processing stage, before storage:

Field class	Control
`created_by.login`, `created_by.name`	**Hash** where the analysis only needs correlation, not identity. Note a hash of a corporate email is enumerable — a control against casual exposure, not a privacy guarantee
`source.item_name`	**Consider masking.** File names leak project names, client names, and deal names
`ip_address`	Keep — needed for the § 11 and § 12 access analysis
`additional_details` on high-volume event types	**Remove.** Cheapest control, apply first

9.3 Bucket isolation

Put Box events in a **dedicated bucket**. Two independent reasons, either sufficient:

- **Retention differs.** An audit trail is typically kept far longer than application logs, and paying application-log retention across 45 GB/month of Box events is expensive.

- **Access differs.** This feed names individuals and the documents they touched. It should be readable by security and compliance, not by everyone who can read application logs. A bucket is the boundary that makes that enforceable — see **ORGNZ**.

> **Sources:**
>
> - Log ingestion limits (DT docs) — *"The event is dropped if the log event contains a timestamp before the current time minus 24 hours."*
> - What's new in Dynatrace SaaS 1.348 (DT docs) — *"Log ingestion now accepts log data up to 72 hours old"*; GA, rollout start 10/12/2026; re-read 10/23/2026
> - Processing in OpenPipeline (DT docs)
> - Configure data storage and retention for logs (DT docs) — *"This means you can extract metrics and generate alerts from records that you won't store."*
> - Log ingestion (DT docs) — the ingestion-route overview this section's API choice sits inside

---

10. Field Mapping Starting Point

Box events are JSON, so the parse is mechanical. What matters is normalizing to names that read the same as the rest of your data.

Box field	Suggested Dynatrace field	Notes
`event_id`	`box.event_id`	**Dedupe key.** Required — streaming repeats events
`event_type`	`box.event_type`	Primary dimension. `LOGIN`, `UPLOAD`, `DOWNLOAD`, `SHARE`, `DELETE`, `SHIELD_ALERT`, …
`created_at`	`timestamp`	Map to the record timestamp, not a custom field
`created_by.login`	`box.user.login`	Consider hashing (§ 9.2). Empty on `SHIELD_ALERT` — use `additional_details.shield_alert.user.email` there (§ 11.2)
`created_by.id`	`box.user.id`	Stable across renames — prefer for joins
`created_by.name`	`box.user.name`	Display only

Box field	Suggested Dynatrace field	Notes
`source.item_type` (file and folder events) or `source.type` (other sources, such as users)	`box.item.type`	File and folder sources carry `item_type` / `item_id` / `item_name`; `type` / `id` appear only on other source objects
`source.item_id` or `source.id`	`box.item.id`	Stable identifier — same split as the row above
`source.item_name`	`box.item.name`	Consider masking (§ 9.2)
`ip_address`	`box.ip_address`	Retain — needed for access analysis
`additional_details`	`box.details.*`	Enterprise-stream only. Shape varies by `event_type`

Normalize before you extract. Metric dimensions and any Smartscape identifiers are computed from fields as they stand at their stage, so renaming after extraction leaves metrics carrying the old dimension names. Rename in Processing, then extract — the same ordering rule as § 9.2.

If you also ingest identity events from elsewhere (Okta, Entra ID), map `box.user.login` onto whatever principal field those feeds use. Correlating a Box download with the sign-in that preceded it is one of the highest-value things this integration enables, and it only works if the principal is spelled the same way in both.

> **Sources:** Event source (Box Dev Docs) — *"The type of the item that the event represents. Can be file or folder."*, Shield Alert Events (Box Dev Docs).

---

11. Box Shield Alerts

Box Shield is Box's own threat detection. If it is licensed, its alerts arrive **in the same event stream** you are already collecting — no second integration, no extra credential. If it is not licensed, no `SHIELD_ALERT` events exist and nothing below applies.

11.1 What Shield detects

Alert type	Trigger
Suspicious Locations	Access from unusual or excluded geographies

Alert type	Trigger
Suspicious Sessions	Unusual user-agents, new IPs, implausible travel
Anomalous Downloads	Download behaviour consistent with data theft
Malicious Content	Malware detected on upload
Ransomware Activity	File extensions consistent with ransomware (**Shield Pro only**)

11.2 The alert payload

Shield alerts follow the standard event schema with `event_type == "SHIELD_ALERT"`, carrying an `additional_details.shield_alert` object:

Field	Use
`rule_category`, `rule_name`	Classification — the primary dimension for routing
`risk_score`	Numeric severity; good threshold for problem creation
`priority`	Box's own severity label
`alert_summary`	Context: IPs, affected files, timestamps
`link`	Deep link back to the Shield console — **put this in the notification**
`user` (`id`, `name`, `email`)	The affected account. The event's own `created_by` is Box's anonymous user (`id` 2, `name` "Unknown User", empty `login`), so route and correlate on this field instead
`rule_response_action`	What Shield already did (e.g. terminated the session)

The `link` field deserves emphasis. An alert that reaches an analyst without a route back to the vendor console costs several minutes of navigation per incident. Carry it through processing into the notification payload.

11.3 Routing them

Shield alerts are the one Box signal that should page someone. Route them per **FAQ-21**:

1. Extract `SHIELD_ALERT` events in OpenPipeline into a **davis event** or business event.
2. Attach ownership so the trigger can route on it — FAQ-21 covers why `dt.owner` on an event needs an explicit, non-retroactive Problem fields mapping.
3. Trigger a **problem-triggered workflow**, filtering on `rule_category` and `risk_score`.

4. **Keep these records.** This is the category where raw retention is justified — § 9.2's discard-by-default posture explicitly exempts it.

```
// Shield alert extraction. Runs standalone against a synthetic record — zero
// bytes scanned. Replace the `data record(...)` line with:
//   fetch logs, from:-24h | filter log.source == "box.events"
// once your collector is running.
// On SHIELD_ALERT, created_by is Box's anonymous user with an EMPTY login —
// the affected account is additional_details.shield_alert.user.
data record(content = "
{\"event_id\":\"a1\",\"event_type\":\"SHIELD_ALERT\",\"created_at\":\"2026-08-
25T10:15:02-07:00\",\"created_by\":{\"type\":\"user\",\"id\":\"2\",\"name\":\"Unknown
User\",\"login\":\"\"},\"additional_details\":{\"shield_alert\":
{\"rule_category\":\"Anomalous Download\",\"rule_name\":\"Bulk
download\",\"risk_score\":78,\"priority\":\"high\",\"alert_id\":9911,\"user\":
{\"id\":2320,\"name\":\"Jane
Doe\",\"email\":\"jdoe@example.com\"},\"link\":\"https://app.box.com/master/shield/al
erts/9911\"}}}")
| parse content, "JSON:evt"
| filter evt[event_type] == "SHIELD_ALERT"
| fieldsAdd box.alert.category   = evt[additional_details][shield_alert]
[rule_category],
            box.alert.risk_score = evt[additional_details][shield_alert][risk_score],
            box.alert.priority   = evt[additional_details][shield_alert][priority],
            box.alert.link       = evt[additional_details][shield_alert][link],
            box.user             = evt[additional_details][shield_alert][user]
[email],
            box.created_by_login = evt[created_by][login]
| fields box.alert.category, box.alert.risk_score, box.alert.priority, box.user,
box.alert.link, box.created_by_login
```

> **Sources:** Shield Alert Events (Box Dev Docs) — the documented sample shows `created_by` as id 2, *"Unknown User"*, with an empty login, and the account under `shield_alert.user`; Box Shield (Box). **Tenant observation (10/02/2026):** the query above returns `box.user` = `jdoe@example.com` and an empty `box.created_by_login`. **Derived:** § 11.3's routing sequence applies FAQ-21's four-axis model to the Shield payload; the recommendation to carry `link` into the notification is an operational inference, not a documented requirement.

---

12. Querying Box Events in Grail

The queries in § 12.1 and § 12.2 run standalone at **zero bytes scanned** — swap the `data record(...)` line for `fetch logs, from:-24h | filter log.source == "box.events"` once data

flows. The queries in § 12.3 and § 12.4 read real logs and are billed as queries; each scanned more than a gigabyte on the validation tenant with no Box data present (10/02/2026). Keep their timeframes short, and scope them to your Box bucket once it exists.

12.1 Parse a raw Box event

If you ingested `content` as the raw JSON (as both collectors in § 7 and § 8 do), this is how you get at the fields at query time. Doing it in OpenPipeline instead is cheaper — see § 9.2 — but this form is useful while validating.

```
// Parse a raw Box event record into queryable fields.
// Runs standalone — zero bytes scanned.
data record(content = "{\"event_id\":\"f82c314e-9a01-4f2b-b6d2-
1c7a55e0d311\",\"event_type\":\"LOGIN\",\"created_at\":\"2026-08-25T10:15:02-
07:00\",\"created_by\":{\"type\":\"user\",\"id\":\"12345\",\"name\":\"Jane
Doe\",\"login\":\"jdoe@example.com\"},\"source\":
{\"type\":\"user\",\"id\":\"12345\"},\"ip_address\":\"203.0.113.42\"}")
| parse content, "JSON:evt"
| fieldsAdd box.event_id   = evt[event_id],
            box.event_type = evt[event_type],
            box.user       = evt[created_by][login],
            box.ip         = evt[ip_address]
| fields box.event_id, box.event_type, box.user, box.ip
```

12.2 Verify deduplication is working

`admin_logs_streaming` returns duplicates by design. At 1M events/day an unnoticed duplicate rate is a direct and recurring cost. Run this weekly.

```
// Duplicate rate on the Box feed. Anything above ~0 means dedupe is not
// working — check that the collector's `seen` set spans the whole run.
// Runs standalone against synthetic records — zero bytes scanned.
data record(box.event_id = "evt-001", box.event_type = "LOGIN"),
     record(box.event_id = "evt-001", box.event_type = "LOGIN"),
     record(box.event_id = "evt-002", box.event_type = "DOWNLOAD"),
     record(box.event_id = "evt-003", box.event_type = "SHIELD_ALERT")
| summarize {ingested = count(), unique = countDistinct(box.event_id)}
| fieldsAdd duplicates = ingested - unique
| fieldsAdd duplicate_pct = round(100.0 * duplicates / ingested, decimals: 1)
```

12.3 Confirm records landed in the right bucket

A bucket misassignment is silent — the data arrives, is queryable, and quietly inherits the wrong retention and the wrong access boundary.

```
// Which bucket are Box records actually in? Expect your dedicated bucket,
// not `default_logs`. An empty result means no Box data yet — not success.
fetch logs, from:-24h
| filter log.source == "box.events"
| summarize records = count(), by:{dt.system.bucket}
| sort records desc
```

12.4 Daily volume against the sizing assumption

Confirm reality matches the § 5 budget. A sustained overshoot is the earliest warning that retention posture needs revisiting.

```
// Daily Box event volume vs. a 1M/day planning assumption.
// Use interval:24h — interval:1d raises a calendar-duration warning.
fetch logs, from:-7d
| filter log.source == "box.events"
| makeTimeseries events = count(), interval:24h
| fieldsAdd daily_avg = arrayAvg(events)
| fieldsAdd pct_of_1m_budget = round(100.0 * arrayAvg(events) / 1000000, decimals: 1)
```

---

13. Operating the Collector

This section is not optional, and it is the one most often skipped.

The failure mode: the collector stops. It produces no logs. No logs is indistinguishable from a quiet Box tenant. Nobody notices. Two weeks later the `admin_logs_streaming` window has rolled past the gap and **those events are gone from the live stream** — the only way back is an `admin_logs` backfill, which reaches back one year (§ 13.3), and past a year they exist only in the Box Admin Console's exported reports, outside the API.

Because Box retains the streaming feed for only two weeks, your detection window for a collector outage is *shorter than two weeks*. Alert on it.

13.1 The staleness query — and the bug in the obvious version

The natural way to write this check has a defect worth seeing, because it fails in the exact direction that hurts.

```
| fieldsAdd collector_status = if(minutes_since_last &gt; 15, "STALLED", else: "OK")
```

When the collector has **never** run, `last_event` is null, so `now() - last_event` is null, `round(null)` is null, and `null > 15` evaluates false — landing on **`"OK"`**. The health check

reports healthy precisely when there is no data at all. Verified against a live tenant on 08/25/2026: this form returned `collector_status: "OK"` alongside `events: 0`.

Test the record count explicitly:

```
// Collector health. Note the explicit events == 0 branch: without it, a
// collector that has NEVER run reports "OK", because null &gt; 15 is false.
// Verified against a live tenant 08/25/2026.
fetch logs, from:-24h
| filter log.source == "box.events"
| summarize {last_event = takeMax(timestamp), events = count()}
| fieldsAdd minutes_since_last = round((now() - last_event) / 1m, decimals: 1)
| fieldsAdd collector_status = if(events == 0, "NO DATA — collector never ran or is
misconfigured",
                            else: if(minutes_since_last &gt; 15, "STALLED",
                            else: "OK"))
```

13.2 What to alert on

Condition	Severity	Why
No Box events for **> 15 min**	High	At a 5-min schedule, three missed runs is a real outage
No Box events **ever** (`events == 0`)	High	Misconfiguration — usually the outbound allowlist, the admin permission, or the app's scopes
Duplicate rate rising above ~0	Medium	Dedupe broken; you are paying twice
Daily volume deviating sharply from baseline	Medium	Either a Box-side change or a partial collection failure
Workflow execution failure	High	Route via **WFLOW** — it is already built in

13.3 Recovering from an outage

Outage length	Recovery
< 2 weeks	Reset the cursor to `0` and re-run. The streaming window still holds the events; dedupe on `event_id` protects against the overlap. Anything older than 24 h lands with its **ingest time** as `timestamp` (log ingest drops older timestamps — § 9.1); query recovered events on `box.created_at`
> 2 weeks	Streaming cannot help. Backfill with `admin_logs` and `created_after` (§ 8.3) — up to one year, chronological and duplicate-free, again with event time carried in `box.created_at` rather than `timestamp`

Outage length	Recovery
Cursor lost, collector healthy	Same as above: pick the stream by how far back the gap goes

Recovery is complete in content, not in time: recovered events keep their real time in `box.created_at`, so time-bucketed dashboards built on `timestamp` show the backlog at the moment it was ingested. The existence of a one-year `admin_logs` backfill path is exactly why § 8's Python collector is worth having even in a Workflow-first design. **Build it before you need it** — writing a backfill tool during an active compliance gap is not when you want to be learning the API.

> **Sources:** Get Enterprise Events (Box Dev Docs), Log ingestion limits (DT docs). **Tenant observation (08/25/2026):** § 13.1's null-comparison behaviour was observed directly by executing both query forms against a live tenant. The Box page also states the retention split behind § 13.3: *"One year of enterprise events are available when stream_type is set to admin_logs."*

---

14. Availability Monitoring

Two different questions, two different answers, and they disagree more often than you would expect.

Question	Source	What it tells you
Is Box up?	`https://status.box.com/api/v2/status.json`	Box's own declaration, per component
Is Box up *for us*?	Dynatrace Synthetic HTTP monitor	Reachability and latency from your locations and networks

14.1 The Box status feed

Box runs a Statuspage instance with a public JSON API — verified reachable and well-formed on 08/25/2026:

Endpoint	Returns
`/api/v2/status.json`	Overall indicator and description

Endpoint	Returns
`/api/v2/summary.json`	Per-component status — Content API, Box Web Application, Login/SSO, Uploads/Downloads, Box Platform / API, and others

`summary.json` is the more useful of the two: "Box is up" is far less actionable than "the Content API is degraded", especially when your integration depends on exactly that component.

Poll it from the same Workflow that collects events — it is one extra `fetch` and requires no authentication.

14.2 Synthetic monitoring — when it adds something

A vendor status page reports what the vendor has decided to declare, on the vendor's timeline. A synthetic monitor reports what your users would actually experience. Add synthetics when:

- Box is on a critical business path and you need independent confirmation before opening a vendor ticket.
- You need per-location or per-network detail — a problem in one office is invisible to a global status page.
- You want Box availability in an SLO (§ 15) and need a signal you control the measurement of.

Point an HTTP monitor at a stable, authenticated-free endpoint and alert on both availability and latency. Do not point synthetics at the Events API — you would be consuming your own rate-limit budget to test a service you are already polling.

> **Sources:** Box status page (Box). **Observation (08/25/2026):** § 14.1's endpoint list and component names were read directly from the live Statuspage API on that date.

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15. Dashboards, Alerting, and SLOs

15.1 Dashboards worth building

Dashboard	Audience	Content
Box security posture	Security	Shield alerts by category and risk score; external shares; failed logins; access from new geographies
Box usage	IT / business	Active users, uploads and downloads by volume, most-active folders, storage trend

Dashboard	Audience	Content
Box integration health	Observability team	Collector freshness (§ 13), duplicate rate, daily volume vs. budget, ingest failures
Your apps ↔ Box	Application teams	Client-span latency and error rate against `box.com` (§ 3.1)

The third one is the one teams forget, and it is the one that protects the other three. A security dashboard fed by a dead collector shows an encouraging absence of alerts.

15.2 What deserves an alert

Signal	Alert?	Notes
`SHIELD_ALERT` above a risk-score threshold	**Yes**	Page security. Route per FAQ-21
Collector stale or absent	**Yes**	§ 13 — highest-consequence operational alert here
Failed-login spike	**Yes**	Credential-stuffing signal
External share of a regulated folder	**Yes**, if you can identify those folders	Otherwise noise
Individual downloads	**No**	Volume without signal. Use Shield's anomaly detection
Box status page degradation	**Maybe**	Informational unless Box is on a critical path

15.3 SLOs

Box is a vendor service — you cannot control its reliability, so an SLO on Box itself is a reporting instrument, not an engineering target. What genuinely belongs in an SLO is **your integration**:

SLO	SLI	Why it is the right target
Audit completeness	% of 5-min windows with at least one successful collection	Directly measures the compliance guarantee you actually made
Collection freshness	% of time the newest event is < 15 min old	The detection window that keeps § 13 honest

The first of those is the one an auditor will ask about. See the **SLO** series for construction.

---

16. Implementation Sequence and Checklist

16.1 Sequence

Phase	Do	Gate before moving on
1. Free signal	Run the § 3.1 span query	Know whether you already have Box performance data
2. Scope	Decide which signal classes you need and why	Written answer to "which questions are we buying?"
3. Box side	Create the app (CCG) with the *Manage enterprise properties* scope, authorize it, grant admin + reports permission	A manual `curl` against `/events` returns entries
4. Dynatrace side	Credential Vault entry, `api.box.com` allowlisted, cursor store tested, bucket created	A test run reaches Box, and a cursor written in one task is read back in the next (§ 7.1)
5. Collector	Build per § 7 (or § 8), run at low frequency	Events arriving in the right bucket (§ 12.3)
6. Processing	OpenPipeline: parse, mask, extract metrics, discard raw	Metrics exist *and* raw volume has dropped
7. Operate	Staleness alert (§ 13), duplicate check, volume check	Alerts proven by deliberately stopping the collector
8. Downstream	Dashboards, Shield routing, SLOs	—

Phase 7 is proven by breaking it on purpose. An untested staleness alert is an assumption, and the cost of that assumption being wrong is a permanent, undetectable audit gap.

16.2 Administrator checklist

Box side

- [ ] Box app created with **Client Credentials Grant**
- [ ] App scope **Manage enterprise properties** checked
- [ ] App authorized in the Box Admin Console (a fresh app is *not* authorized by default) — and re-authorized after any scope change
- [ ] Service account is enterprise admin/co-admin with **"Run new reports and access existing reports"**

- [ ] Confirmed `GET /events?stream_type=admin_logs_streaming` returns entries with a manual call
- [ ] Shield licensing confirmed (or § 11 knowingly skipped)

Dynatrace side

- [ ] `client_id` / `client_secret` in the **Credential Vault** — not in workflow code — with **AppEngine** scope, **Allow access without app context** on, and the workflow actor granted access
- [ ] **`api.box.com` under Settings > General > External requests** (Workflow path only; add `status.box.com` too if you poll it)
- [ ] Cursor store verified from a workflow: `setAppState` in one task, `getAppState` in the next (§ 7.1)
- [ ] Collector is a **standard** workflow, not simple
- [ ] Dedicated bucket created with agreed retention
- [ ] Platform token with `openpipeline:logs:ingest` — or, on a classic or hybrid environment, a classic token with `logs.ingest` (Python path only)
- [ ] OpenPipeline: parse → mask/hash → extract metrics → **`No storage assignment`** in the Bucket assignment stage, never `Drop record` in Processing

Operations

- [ ] Cursor persisted, **no TTL**, written only after successful ingest
- [ ] Staleness alert live **and tested by stopping the collector**
- [ ] Duplicate-rate check scheduled
- [ ] Backfill path (§ 8) built and tested *before* it is needed — including that events older than 24 h arrive (ingest-time `timestamp`, real time in `box.created_at`)
- [ ] Retention and masking signed off by whoever owns the data

---

17. Common Gotchas

#	Gotcha	Symptom	Fix
1	**Reaching for webhooks**	Receiver works; the events you need never arrive	Webhooks are file/folder-scoped and carry no audit events. Poll `/events` (§ 2.2)
2	**Host not on the outbound allowlist**	The task fails on the first `fetch`, or the run is empty if the code catches errors	Add `api.box.com` under **Settings > General > External requests**

#	Gotcha	Symptom	Fix
3	**Missing the reports permission or the app scope**	API returns no entries, or an authorization error	Grant "Run new reports and access existing reports" to the service account, **and** check the app's *Manage enterprise properties* scope (§ 4.1)
4	**TTL on the cursor**	Collector silently resets weeks later	`stateClient` TTL caps at 90 days. Omit `validUntilTime`
5	**Cursor written before ingest**	Events vanish on a failed ingest	Write the cursor only after ingest succeeds (§ 7.3)
6	**Not deduplicating**	Inflated counts and ingest cost	`admin_logs_streaming` repeats events. Dedupe on `event_id` (§ 12.2)
7	**Assuming a URL field is populated**	Zero rows, reads as "no Box traffic"	`url.full` (`stable`) and `http.url` (not in the dictionary) can each be empty on a tenant. Key on `server.address`; check `countIf(isNotNull(…))` before relying on a URL field (§ 3.1)
8	**Staleness check returns "OK" with no data**	Dead collector reports healthy	`null > 15` is false. Test `events == 0` explicitly (§ 13.1)
9	**`Drop record` in the Processing stage**	Raw discarded *and* metrics never built	Use `No storage assignment` in the Bucket assignment stage (§ 9.2)
10	**`box-sdk-gen` instead of `boxsdk` — or `from boxsdk import …`**	Building on a deprecated SDK, or `ModuleNotFoundError` on v10	Install `boxsdk` v10+, import from `box_sdk_gen` (§ 8.1)
11	**Simple workflow**	No **Run JavaScript** action is available	Use a standard workflow (§ 7.1)
12	**Backfill sharing the live cursor**	Live collector jumps backwards or forwards	Separate cursor per stream type (§ 8.3)
13	**Renaming fields after extraction**	Metrics carry stale dimension names	Normalize in Processing, then extract (§ 10)
14	**413 on ingest**	Batch inside 10 MB still rejected	Payload expands during processing; ceiling is 16 MB. Halve the batch (§ 9.1)
15	**`interval:1d`**	Calendar-duration warning	Use `interval:24h` (§ 12.4)

#	Gotcha	Symptom	Fix
16	**Backfill stamped with event time**	Replay "succeeds"; events older than 24 h are silently dropped (Python sees a **200**, not an error)	Omit `timestamp` for old events; carry `box.created_at` (§ 9.1)
17	**Testing `e.status` on a client-state error**	First Workflow run always throws	Use `isNotFound(e)` from `@dynatrace-sdk/client-state` (§ 7.3)
18	**Credential missing "Allow access without app context"**	The first run fails reading the credential	Workflow tasks have no app context — turn the setting on and grant the workflow actor access (§ 7.1)
19	**Shield alerts routed on `created_by.login`**	Every alert has an empty user	Use `additional_details.shield_alert.user` (§ 11.2)

Gotchas 1, 7, and 8 share a shape worth naming: **each produces a confident, plausible, wrong answer rather than an error.** A webhook that delivers nothing, a span query that returns zero, and a health check that says OK all look like success. Every one of them was reproduced during the research for this entry.

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18. Recommended Approach

Query your own spans first. If your services call Box, that data exists now and costs nothing (§ 3.1). It may answer the question that prompted the project.

Then build one collector, in a Workflow. At 1M events/day a Workflow clears the 120-second ceiling with roughly 90% headroom, keeps secrets in the Credential Vault, and inherits scheduling and failure alerting. Poll `admin_logs_streaming` every 5 minutes, dedupe on `event_id`, persist the cursor with `stateClient` and no TTL — after confirming in a test workflow that app state is reachable from a Run JavaScript task (§ 7.1) — and write that cursor only after a successful ingest.

Write the Python backfill before you need it. It is the only recovery path for an outage longer than two weeks, and the worst time to write it is during one. It recovers the events, not their place on the timeline: log ingest drops timestamps older than 24 h, so recovered events carry their real time in `box.created_at`.

Decide retention before switching on. At ~45 GB/month, extract-then-discard is not an optimization — it is the difference between a proportionate cost and an unpleasant surprise. Keep Shield alerts and regulated-folder activity; discard the `PREVIEW` and `DOWNLOAD` bulk once a counter has captured it.

Alert on the collector, and prove the alert works by stopping it. Everything else in this entry protects data you have. This is the only control that protects data you have not collected yet — and Box's two-week streaming window means an unnoticed outage becomes permanent.

What "done" looks like: Shield alerts routed to security with a working console link; a security dashboard whose freshness you can vouch for; audit records in a bucket with retention and access someone signed; and a staleness alert that has been tested by breaking it on purpose.

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